

Deep Generative Model for Image-Space 3D Microstructure Synthesis of UD Fiber-reinforced Polymer Composites

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Introduction

The current pipeline converts CT cross-sections into a set of hand-engineered statistical descriptors (fiber centroids, ellipse axes, tilt angles $\theta_x, \theta_y, \theta_z$) and then explicitly models their joint distribution. This decoupled representation is powerful but makes assumptions: fiber morphology is reduced to an ellipse, packing density is encoded through a copula, and depth correlation is captured by a sliding-window Gaussian model. A complementary approach is to treat each depth slice directly as an image and to learn the structure of the microstructure (fiber size distribution, spatial clustering, neighbor packing, and how all of these evolve across depth) entirely from data, without manually specifying a statistical model.

Aim: Develop and validate a depth-conditioned generative model such as Denoising Diffusion Probabilistic Models (DDPMs), Conditional Variational Autoencoders (CVAEs), etc. that learns the 3D fiber microstructure directly from stacked cross-sectional synthetic CT images, and evaluate whether it reproduces the spatial statistics captured by the current analytical statistical pipeline and the original CT scans. Including copula inter-slice correlation, spatial neighborhood persistence, and per-fiber misalignment distributions.

Research Question: Can a depth-conditioned deep generative model trained on stacked cross-sectional CT images of CFRP microstructures reproduce the inter-slice statistical coupling (copula ρ , neighborhood persistence, etc.) at a fidelity equivalent to or exceeding the explicit Gaussian model, without manual feature engineering?

Activities & Expectations

1. Literature review on generative models in microstructure synthesis: GANs, DDPMs, CVAEs, and video diffusion (relevant for slice-stack continuity); benchmark on 2D microstructure tasks.
2. Build a depth-conditioned DDPM or CVAE where the latent conditioning vector encodes depth z and packing density.
3. Validation of the pipeline and generated microstructure against the ground truth.
4. Sensitivity study: Evaluate how many CT scans (training examples) are needed for generative quality to saturate, relevant to practical deployment in industrial settings with limited data, or expensive few CT scans.

Expected outcome: A generative model that can, given a target packing fraction and depth range, produce a statistically valid 3D microstructure cross-section stack. The result is the quantified comparison of fidelity/accuracy vs the current analytical pipeline.

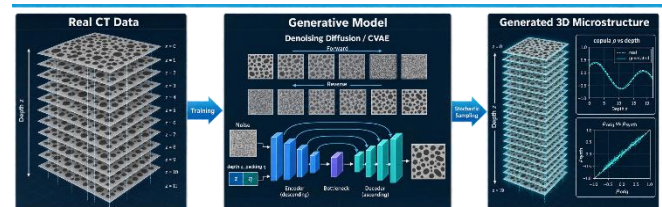


Figure 1. Preliminary pipeline, [fig made by ChatGPT].

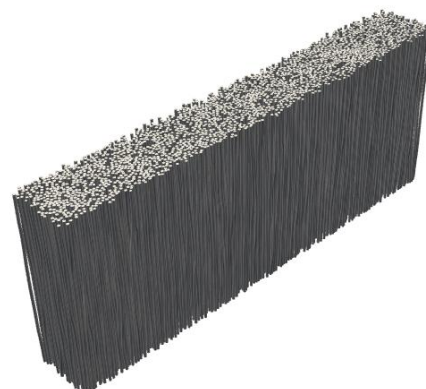


Figure 2. Current pipeline used to generate synthetics statistically equivalent microstructure from Xray-CT scans.

